CommunityFix

Evaluation of prototype 1, built by Firmware Force.

Introduction:

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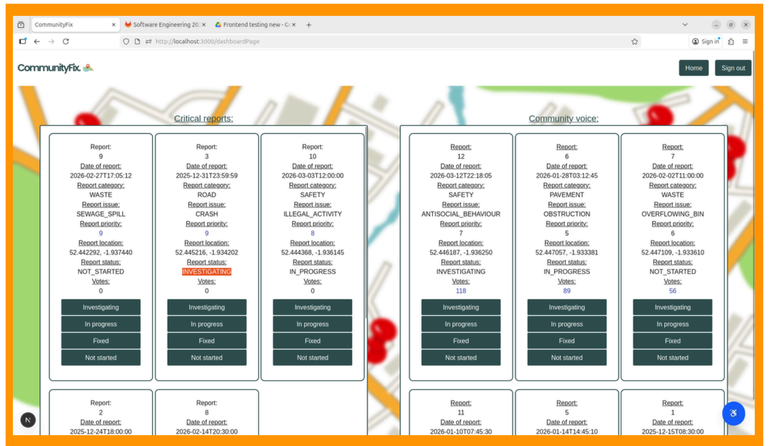
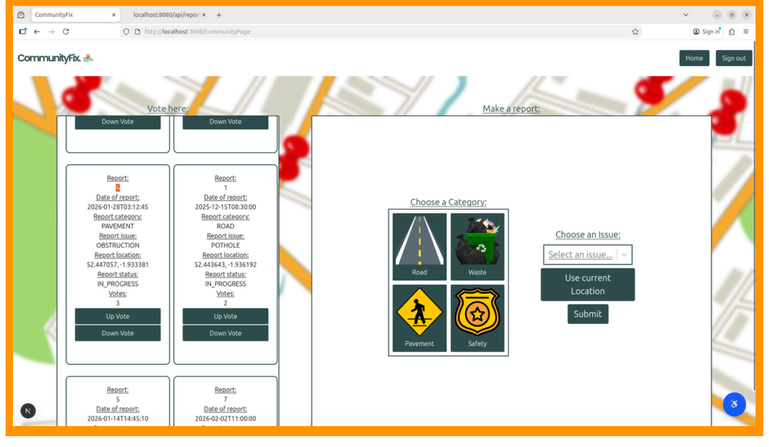
Make cities and human settlements inclusive, safe, resilient and sustainable

11.3: By 2030, enhance inclusive and sustalinable urbanization and capacity for participatory integrated and sustainable human settlerent planning and management.
11.6: By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management.

Our prototype aims to tackle SGD 11.3/11.6 of the united nation's sustainable development goals by enhancing human settlement management and wast reduction. This product allows the community to report issues facing them to the council in an efficent and simple way, and allows the council to view the issues ordered by priority on a modern and easy to use dashboard so that the authorities can manage and fix the issues.

1 Our first prototype for CommunityFix aims to tackle SDG 11.3/11.6. The software itself doesn't improve infacstructure, it targets the information gap between localt residents and the council. We believe the app works well to target this, the product provides a direct link from the resident experiencing issues and council who can action on information with votes from the locals. This is more than just collecting important data, it helps voice local opinion with the voting system and keeping priority in mind. The prototype does have limitations as we can't facilitate assignment of a fix but this is a goal for prototype 2. Our team made no significant changes to the original proposed project / product. Our team had numerous successes and a few difficulties along the way. The main success was maintaining a forwards momentum and continued progress by working on different aspects of the full-stack architecture simultaneously. This was done with consistency, regular in-person meetings twice a week and clearly definined tasks for the frontend and the backend. This allowed us to overcome some difficulties the primary challenge we faced was a slow start due to learning frameworks. We faced a steep learning curve as 6 of the 7 team members were unfamiliar with React and Spring-boot and have never undertook a project like this before.

2 We successfully fully implemented the following features: F-2: Accessibility, F-3: Role based access control, F-6: Efficient reporting mechanism, F-7: GPS pinpointing system, F-9: Clear labelling and categorisation of reports and issues, F-10: Status updates on reports, F-11: Text and Icon based system, F-12: Community voting platform, F-13: Database management System, F-14: Centralised dashboard, F-16: Optional anonymous reporting, and we partially implemented F-17: Resident notification on status updates. We did not implement: F-1: Report Filtering and list generation which was only implemented on the backend, F-4: Map displaying live reports and hotspots, F-5: User trust level / reward system, F-8: Public user accounts partially implemented with non-public user accounts, F-15: task allocation system, F-18: Accessible map views, F-19: Offline mode. We didn't implement these features because we believed they did not attribute to core functionality which we were aiming to achieve within the 3 week development sprint. One thing we decided to change was that instead of optional anonymity stated in F-16 all reports should be anonymous and only admins should be able to see the userId of the user who makes the report. The features we would focus on in our second prototype would be F-4/18 the features to build our map, F-5 the user trust levels and reward system, F-8 the public user accounts that show trust levels, and F-15 the task allocation system. We would focus on these features as they add to the USP of the product.

3 We believe we made a good choice regarding what features to focus on for our first prototype we kept a specific set of features chosen to allow us to build core functionality, and kept us grounded within our groups capability, ensuring we did not take on too much for the timeframe. We believe our choice of technologies to be used were appropriate, Using spring boot allowed us to keep the backend in a language we have used before, Using both react and spring boot allowed us to seperate the workload easier and allowed us to keep the front and backend code seperate, The technologies used allowed us to learn industry standard technology. We ended using tailwind css for the software styling improving readability / future maintainability of the front end. During the development process we ended up deviating from the chosen high level class diagrams to remove redundant code like voteReport, Category, and Issue, which we replaced using a voteCount in the report class, and connecting the report class straight to the category and issue enums, to improve logic flow we added the userRole / Status enums. In the next sprint we decided something to improve on would to be to change how we split the team from frontend / backend to splitting on features to improve efficiency. One thing we would adjust is to have designated roles in the team like Project manager, team leader, or team member.

4 To coordinate our work we initially split into 2 teams one focusing on the frontend and one focusing on the backend. Prosper was organising the front end and Elliot was organising the backend. Throughout the process our approach changed with responsiblities and tasks being allocated based on features and components to implement. Some challenges we faced throughout the process were a steep learning curve in learning how to build the front and back end in the react and spring boot frameworks. Another challenge for some of the group was knowing when to stop preparing / designing and to actually start coding and building the software. Some successes we had in the progress was our time management, project management, and teamwork. During the process we also worked on but did not finish an iimag upload to go with the reports, we got it built on the front end but ran into some difficulty on the backend so we decided to complete this addition in prototype 2.